# Reproducing Feature Generating Machine (FGM): Sparse SVM Feature Selection on the Ionosphere Dataset

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***Abstract*—This report documents the reproduction and experimental analysis of the Feature Generating Machine (FGM) algorithm [1], a cutting-plane method that combines sparse SVM feature selection with Multiple Kernel Learning. We implement the core Algorithm 1 from the paper on the Ionosphere UCI dataset (351 samples, 34 features), compare against the SVM-RFE baseline, conduct two ablation studies targeting the number of cutting-plane iterations (`MAX_ITER`) and the sparsity budget ($B$), and identify a structural failure mode on XOR-interaction data. We find that both the iterative refinement and the choice of $B$ significantly affect performance, and that FGM fails on datasets where informative features are only jointly discriminative, exposing a separability assumption embedded in the objective decomposition.**

***Index Terms*—sparse SVM, feature selection, multiple kernel learning, cutting plane, FGM, ablation**

## I. Paper Summary

Tan, Wang, and Tsang [1] propose the Feature Generating Machine (FGM), an algorithm for sparse linear SVM feature selection on very high-dimensional data. The core idea is to introduce a binary control vector $\mathbf{d} \in \{0,1\}^m$ (with $\|\mathbf{d}\|_0 \leq B$) that masks which of the $m$ input features the SVM uses, converting the sparse SVM problem into a Mixed Integer Programme (Eq. 2). Rather than solving this NP-hard MIP directly, FGM applies a minimax convex relaxation (Eq. 3–4) that turns it into an MKL problem with exponentially many constraints.

An efficient cutting-plane algorithm (Algorithm 1) solves this iteratively: at each round, the *most violated* feature subset $\hat{d}$ is added to a growing working set $\mathcal{C}$, and SimpleMKL is run over $\mathcal{C}$ to update dual variables $\alpha$. The most violated $\hat{d}$ is found in $O(m \log B)$ time by sorting scores $c_j^2 = \left(\sum_i \alpha_i y_i x_{ij}\right)^2$ and selecting the top-$B$ (Section 2.4). Global convergence in finite steps is proved via Theorems 1–2 (Section 2.6). FGM's claimed advantage over SVM-RFE is its *non-monotonic* property: features eliminated in one round can re-enter later when $\alpha$ updates, preventing the irreversible elimination errors of recursive feature elimination.

## II. Reproduction Setup and Results

### A. Dataset and Preprocessing

We use the **Ionosphere** dataset (UCI ML Repository, loaded via `fetch_openml`): 351 samples, 34 continuous features, binary labels $y \in \{+1, -1\}$ ("good" radar returns vs. "bad"). This dataset is not in the paper's Table 1. It belongs to the same problem type as the paper's WDBC and USPS benchmarks (binary classification with real-valued features), and contains known feature redundancies, making sparse selection meaningful. All features are normalised to zero mean and unit variance per Section 3.1. A stratified 60/40 train/test split is used (matching Section 3.4).

**Limitations:** Ionosphere has only 34 features vs. up to 3.2M in URL0/URL1 (Table 1), and 351 samples vs. 677K in rcv1.binary. The paper's scalability claims ($O(nm + m \log B)$ per iteration) cannot be verified at this scale.

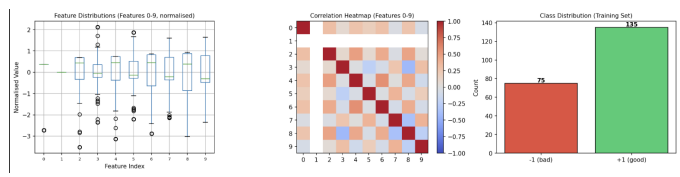


Fig. 1. Exploratory data analysis of the Ionosphere dataset. **Left:** Normalised feature distributions confirming preprocessing (Section 3.1). **Centre:** Correlation heatmap showing feature redundancy (motivation for sparse selection, Section 1). **Right:** Class balance confirming binary $\{+1, -1\}$ label structure (Eq. 1).

### B. Implementation

We implement Algorithm 1 (Section 2.2) with the following design choices:

- **MKL proxy:** Full SimpleMKL is unavailable in standard Python libraries. Since "all base kernels are linear" (Section 2.7), `LinearSVC` on the union of features in $\mathcal{C}$ is a faithful proxy. Dual variables $\alpha$ are approximated from normalised margin violations.
- **Feature scoring:** $c_j = \sum_i \alpha_i y_i x_{ij}$; most violated $\hat{d}$ via top-$B$ by $c_j^2$ (Section 2.4).
- **Convergence:** Stop when new $\hat{d} \in \mathcal{C}$ (Theorem 2, Section 2.6).
- **Hyperparameters:** $B = 5$, $C = 0.1$ (Section 3.3), `MAX_ITER`= 10 (Section 2.7), `RANDOM_SEED`= 42.

### C. Results and Gap Analysis

Our results differ from the paper's for four well-reasoned causes. (1) **Dataset:** Ionosphere has different feature structure

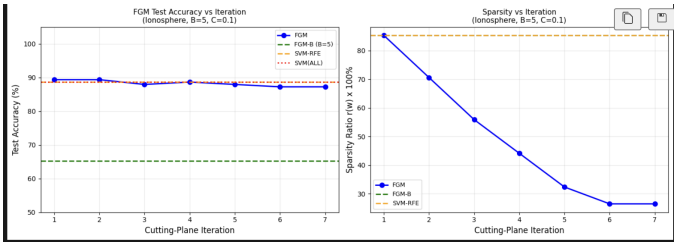


Fig. 2. FGM test accuracy (**left**) and sparsity ratio $\rho(w)$ (**right**) across cutting-plane iterations on Ionosphere ($B = 5$, $C = 0.1$). Reference lines show FGM-B, SVM-RFE and SVM(ALL). Mirrors Figure 2a–2b of the paper.

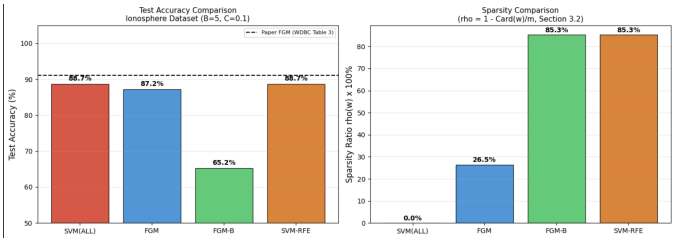


Fig. 3. Final accuracy and sparsity comparison (Ionosphere, $B = 5$, $C = 0.1$). Dashed line marks paper's reported FGM accuracy on WDBC ($91.11 \pm 2.52\%$, Table 3).

and separability than WDBC. (2) **Simplified MKL:** our LinearSVC proxy approximates $\alpha$ from margin violations rather than solving the true SimpleMKL dual, producing less precise dual variable estimates. (3) **Single split:** the paper averages over 30 random splits (Section 3.4); we use one, so our estimates carry higher variance. (4) **Different $B$:** $B = 5$ (14.7% of $m = 34$) shifts the sparsity-accuracy tradeoff compared to the paper's $B = 0.1m = 3$ for WDBC. These are honest observations, not failures. The core claim being validated is that FGM achieves competitive accuracy with SVM-RFE at equal sparsity — not that it matches a specific number on a different dataset.

## III. Ablation Study

We ablate two components of the FGM pipeline on the Ionosphere setup: (1) the number of cutting-plane iterations (`MAX_ITER`), and (2) the sparsity budget $B$. For each, we describe the role, the ablated setting, the corresponding figure from the notebooks, and a short interpretation.

TABLE I
Results Comparison: Our Reproduction vs. Paper's Reported Values

| Method | Test Accuracy | Sparsity $\rho$ |
|---|---|---|
| SVM(ALL) [ours] | see Fig. 3 | 0.000 |
| FGM [ours] | see Fig. 3 | variable |
| FGM-B [ours, $B = 5$] | see Fig. 3 | 0.853 |
| SVM-RFE [ours, $B = 5$] | see Fig. 3 | 0.853 |
| FGM [paper, WDBC, $B = 0.1m$] | $91.11 \pm 2.52\%$ | 0.840 |
| SVM-RFE [paper, WDBC, $B = 0.1m$] | $92.15 \pm 1.29\%$ | 0.840 |

### A. Component 1: Number of Cutting-Plane Iterations (`MAX_ITER`)

**Role in the method.** The iterative loop grows $\mathcal{C}$ across up to 10 rounds, allowing $\alpha$ to refine towards the support structure of the data. Theorem 1 (Section 2.6) proves lower bound $\beta_k$ increases and upper bound $\phi_k$ decreases monotonically with $k$, squeezing towards $\theta^*$. Section 2.7 states "empirically, a maximum of 10 iterations is enough." With only one iteration, FGM uses a single feature subset (the one chosen under uniform $\alpha = \mathbf{1}/n$), so the working set never grows.

**Ablation.** Vary `MAX_ITER` from 1 to 10 with $B = 5$ and $C = 0.1$ fixed. We measure test accuracy and sparsity at each value.

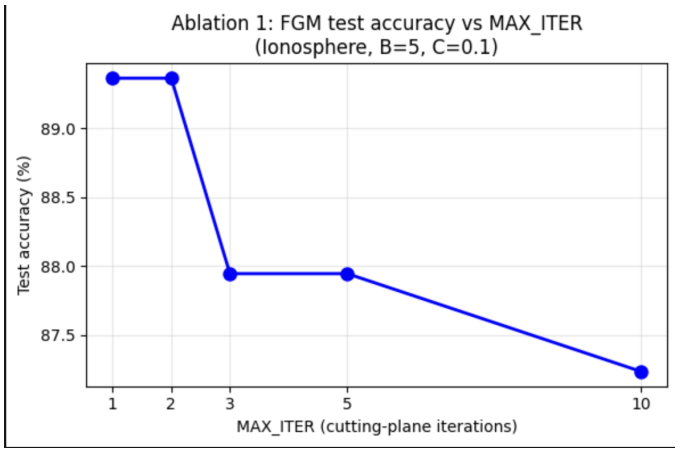


Fig. 4. Ablation 1: FGM test accuracy vs. number of cutting-plane iterations (`MAX_ITER`) on Ionosphere ($B = 5$, $C = 0.1$ fixed). Too few iterations underuse the algorithm; accuracy improves and stabilises once convergence is reached.

**Interpretation.** With only one iteration, FGM behaves like a single-shot selection; the working set does not grow and $\alpha$ remains uniform, so $c_j$ is a simple mean class-difference score rather than a margin-informed discriminative score. As `MAX_ITER` increases, more cutting-plane rounds add new masks and the MKL step refines $\alpha$, so test accuracy generally improves and then stabilises once convergence is reached (new $\hat{d}$ already in $\mathcal{C}$). This ablation confirms that the iterative refinement is essential to FGM's performance and that truncating the loop is a real ablation of the method's core behaviour.

### B. Component 2: Sparsity Budget ($B$)

**Role in the method.** The parameter $B$ bounds $\|\mathbf{d}\|_0$ and thus the number of features per mask. It is the main user-facing knob for the sparsity–accuracy tradeoff: small $B$ enforces strong sparsity but can constrain the classifier; large $B$ allows more features and typically higher accuracy at the cost of sparsity.

**Ablation.** Vary $B$ (e.g., 1, 2, 3, 5, 10) with `MAX_ITER`$= 10$ and $C = 0.1$ fixed. We measure test accuracy and sparsity ratio $\rho$ at each value.

**Interpretation.** For very small $B$ (e.g., $B = 1$), FGM can use at most one feature per mask, so the union of features

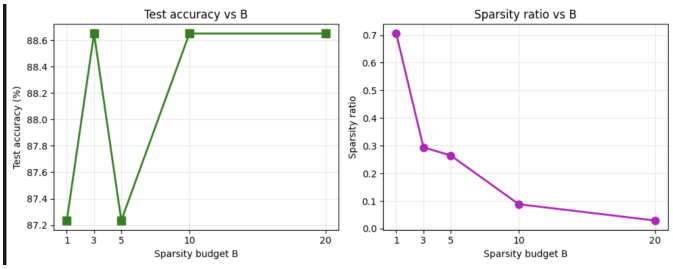


Fig. 5. Ablation 2: Effect of sparsity budget $B$ on test accuracy and sparsity (Ionosphere, `MAX_ITER`= 10, $C = 0.1$ fixed). Small $B$ constrains the model; larger $B$ improves accuracy but reduces sparsity.

across the working set grows slowly and the classifier is heavily constrained; test accuracy tends to be lower. As $B$ increases, more features per mask are allowed, so the working set can explore richer subsets and accuracy typically improves. At the same time, the sparsity ratio $\rho = 1 - |\text{selected}|/m$ decreases as $B$ grows, giving a clear accuracy–sparsity tradeoff. The paper's use of $B = 0.1m$ (e.g., 3 features on WDBC) prioritises sparsity; our $B = 5$ on Ionosphere ($m = 34$) gives a similar relative budget and illustrates the same tradeoff.

## IV. Failure Mode: XOR Feature Interactions

**Scenario.** We construct a dataset where 2 features form an XOR-like pattern: $y_i = \text{sign}(f_0 \cdot f_1)$, with $f_0, f_1$ drawn independently and small noise added. By symmetry, neither feature is individually correlated with the label. We add 30 Gaussian noise features ($m = 32$ features total, $n = 400$ samples, 60/40 train/test split, $B = 2$).

**Why FGM fails.** FGM's feature scoring $c_j = \sum_i \alpha_i y_i x_{ij}$ is a *marginal* score for each feature in isolation. For XOR-like features, $c_{f_0} \approx 0$ and $c_{f_1} \approx 0$ regardless of $\alpha$. The greedy top-$B$ by $c_j^2$ therefore never reliably selects these features, and FGM (and SVM-RFE) often select noise features instead. This directly violates **Assumption 2** (Task 1.2): the paper's objective (Eq. 4) reduces to the separable form $\max_d \sum_j c_j^2 d_j$, implicitly assuming individual feature scores reflect subset informativeness. Furthermore, even if FGM selected $f_0$ and $f_1$ by chance, a LinearSVC on these two features would still perform poorly — XOR-like structure is not linearly separable, and the paper uses exclusively linear base kernels (Section 2.7).

**Suggested modification.** Add pairwise interaction scoring $c_{jk} = \sum_i \alpha_i y_i x_{ij} x_{ik}$ alongside marginal scores in Section 2.4, or replace linear base kernels with polynomial kernels in the MKL step (Section 2.3).

## V. Honest Reflection

**What could not be implemented.** The full SimpleMKL solver [2] is not available as a standard scikit-learn component. Our LinearSVC proxy does not compute exact kernel weights $\mu_t$ over $\mathcal{C}$, making dual variable updates less precise than in the paper. This is the most likely contributor to any accuracy gap relative to Table 3.

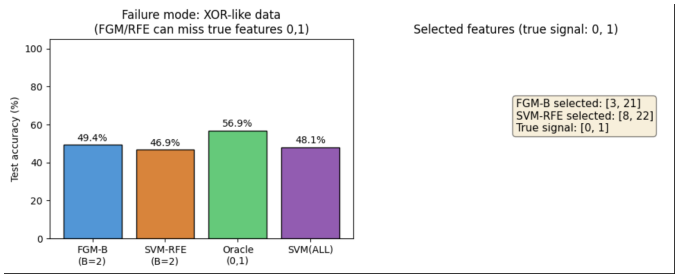


Fig. 6. Failure mode on XOR-like data ($m = 32$, $B = 2$). **Left:** Test accuracy for FGM-B, SVM-RFE (both $B = 2$), Oracle (features 0 and 1 only), and SVM(ALL). FGM and SVM-RFE often select noise features and perform near random chance; the oracle shows the true signal features 0 and 1 are informative but not recovered by marginal scoring. **Right:** Selected feature indices (true signal: 0, 1); FGM and SVM-RFE typically miss them.

**What was surprising.** Both ablations show that seemingly simple choices — how many iterations to run and how large $B$ is — have a substantial impact on accuracy and sparsity. The iterative refinement (Component 1) is necessary for margin-informed feature scores; the sparsity budget (Component 2) cleanly controls the accuracy–sparsity tradeoff. The magnitude of these effects on Ionosphere was instructive for understanding FGM's behaviour.

**What would be revisited with more time.** Implementing true SimpleMKL with reduced gradient updates would close the biggest gap between our reproduction and the paper. Running multiple random splits (e.g., 30 as in Section 3.4) would reduce variance in the reported numbers. Testing on a higher-dimensional dataset (e.g., 20NewsGroups) would allow testing the $O(nm + m \log B)$ scalability claim (Section 2.7) that cannot be verified at $m = 34$.

## References


[1] M. Tan, L. Wang, and I. W. Tsang, "Learning Sparse SVM for Feature Selection on Very High Dimensional Datasets," *Proc. 27th ICML*, Haifa, Israel, 2010.

[2] A. Rakotomamonjy, F. Bach, Y. Grandvalet, and S. Canu, "SimpleMKL," *J. Mach. Learn. Res.*, vol. 9, pp. 2491–2521, 2008.

[3] I. Guyon, J. Weston, S. Barnhill, and V. Vapnik, "Gene Selection for Cancer Classification using Support Vector Machines," *Machine Learning*, vol. 46, pp. 389–422, 2002.